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$^{40}_{19}\mathrm{K}_{21}$

1 Decay Scheme

^{40}K is a natural isotope with an isotopic abundance of 0.011668 (8) %. It disintegrates by β^- emission to the ground state of ^{40}Ca at 89.56 (7) %, by electron capture to the 1460-keV level of ^{40}Ar at 10.34 (7) %, and to the ground state of ^{40}Ar by electron capture at 0.098 (25) % and by β^+ at 0.00103 (13) %.

Le ^{40}K est un isotope naturel dont l'abondance est de 0,011668 (8)%. Il se désintègre à 89,56 (7) % par émission β^- vers le niveau fondamental du ^{40}Ca, par capture électronique à 10,34 (7) % vers le niveau à 1460 keV du ^{40}Ar, et vers le niveau fondamental du ^{40}Ar à 0,098 (25) % par capture électronique et à 0,00103 (13) % par émission β^+.

2 Nuclear Data

$T_{1/2}(^{40}\mathrm{K})$	:	1,2522	(27)	10^9 a
$Q^+(^{40}\mathrm{K})$	:	1504,40	(6)	keV
$Q^-(^{40}\mathrm{K})$	:	1310,91	(6)	keV

2.1 Electron Capture Transitions

	Energy (keV)	Probability (%)	Nature	log ft	P_K	P_L	P_M
$\epsilon_{0,1}$	43,55 (6)	10,34 (7)	Unique 1st Forbidden	11,54	0,7537 (7)	0,22248 (33)	0,02379 (32)
$\epsilon_{0,0}$	1504,40 (6)	0,098 (25)	Unique 3rd Forbidden	21,64	0,8908 (7)	0,0965 (5)	0,01271 (15)

2.2 β^- Transitions

	Energy (keV)	Probability (%)	Nature	log ft
$\beta^-_{0,0}$	1310,91 (6)	89,56 (7)	Unique 3rd Forbidden	20,59

2.3 β^+ Transitions

	Energy (keV)	Probability (%)	Nature	log ft
$\beta^+_{0,0}$	482,40 (6)	0,00103 (13)	Unique 3rd Forbidden	21,64

2.4 Gamma Transitions and Internal Conversion Coefficients

	Energy (keV)	$P_{\gamma+ce}$ (%)	Multipolarity	α_K (10^{-5})	α_L (10^{-5})	α_M (10^{-5})	α_T (10^{-5})	α_π (10^{-5})
$\gamma_{1,0}$(Ar)	1460,851 (6)	10,34 (7)	E2	2,63 (4)	0,215 (3)	0,0210 (3)	10,28 (15)	7,41 (11)

3 Atomic Data

3.1 Ar

ω_K : 0,1199 (28)
$\bar{\omega}_L$: 0,00147 (30)
n_{KL} : 1,697 (6)

3.1.1 X Radiations

		Energy (keV)	Relative probability
X_K			
	$K\alpha_2$	2,95566	50,49
	$K\alpha_1$	2,95774	100
	$K\beta_1$	3,1905	16,24
X_L			
	$L\ell$	0,2195	
	$L\eta$	0,2215	
	$L\beta$	0,3112 - 0,3114	

3.1.2 Auger Electrons

	Energy (keV)	Relative probability
Auger K		
KLL	2,511 - 2,669	100
KLX	2,831 - 2,942	21,6
KXY	3,149 - 3,174	1,16
Auger L	0,17 - 0,31	

4 Electron Emissions

		Energy (keV)	Electrons (per 100 disint.)
e_{AL}	(Ar)	0,17 - 0,31	2,307 (16)
e_{AK}	(Ar)		
	KLL	2,511 - 2,669	6,94 (6)
	KLX	2,831 - 2,942	
	KXY	3,149 - 3,174	
$ec_{1,0\ T}$	(Ar)	1457,645 - 1460,835	0,001063 (17)
$\beta^+_{0,0}$	max:	482,40 (6)	0,00103 (13)
	avg:	251,757 (28)	
$\beta^-_{0,0}$	max:	1310,91 (6)	89,56 (7)
	avg:	582,898 (26)	

5 Photon Emissions

5.1 X-Ray Emissions

		Energy (keV)	Photons (per 100 disint.)	
XL	(Ar)	0,2195 - 0,3114	0,0034 (7)	
$XK\alpha_2$	(Ar)	2,95566	0,286 (8)	$K\alpha$
$XK\alpha_1$	(Ar)	2,95774	0,567 (14)	
$XK\beta_1$	(Ar)	3,1905	0,092 (4)	$K'\beta_1$

5.2 Gamma Emissions

	Energy (keV)	Photons (per 100 disint.)
$\gamma^{\pm}$	511	0,00359 (25)
$\gamma_{1,0}$(Ar)	1460,822 (6)	10,34 (7)

6 Main Production Modes

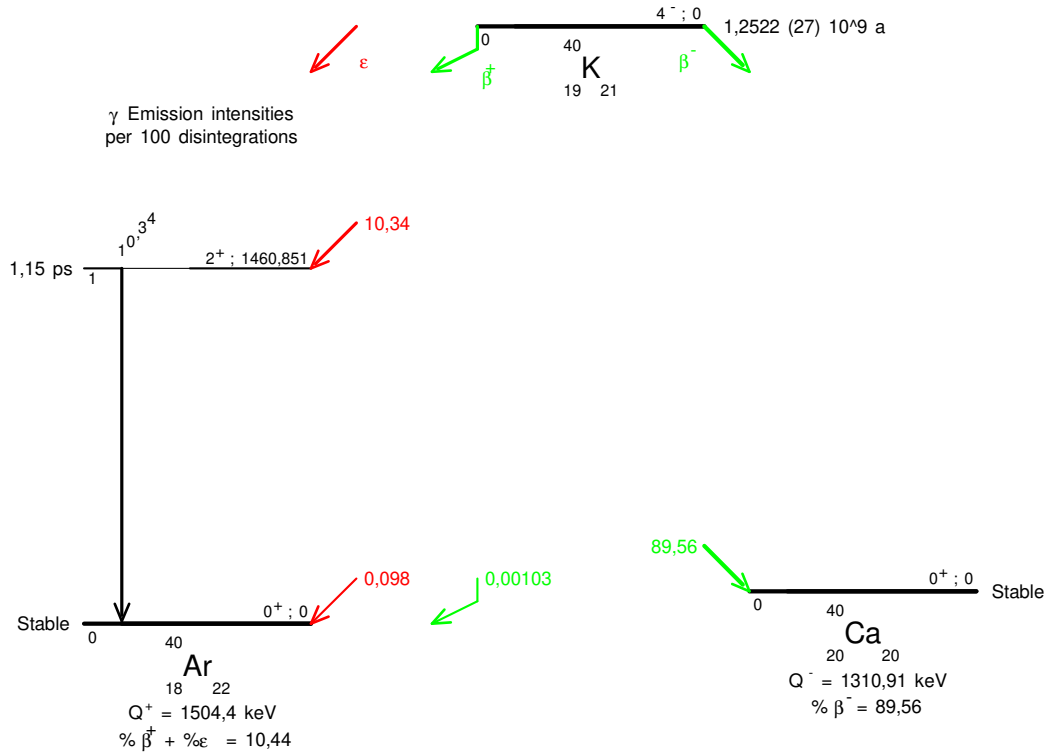
Naturally occurring

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  (m_e)


4⁻ ; 0
1,2522 (27) 10^9 a
ε
β⁺
β⁻
0
40
K
19 21
γ Emission intensities
per 100 disintegrations
10,34
2⁺ ; 1460,851
1,15 ps
1
0,098
0,00103
0⁺ ; 0
Stable
0
40
Ar
18 22
Q⁺ = 1504,4 keV
% β⁺ + %ε = 10,44
89,56
0⁺ ; 0
Stable
0
40
Ca
20 20
Q⁻ = 1310,91 keV
% β⁻ = 89,56